Marking Scheme

SET-A55 · Paper 1 · 105 marks

FOR REVISION

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating：**3.5** / 5.0

General Rules
- M / A / PROOF mark types
- **f.t.**：跟進分：接受由前一步錯誤答案跟進而來的正確運算
- Deduction cap：A(1)=1 · A(2)=1 · B=1 (u-1, pp-1)
- u 及 pp 於每部（甲部(1)、甲部(2)、乙部）各最多扣 1 分
- M：方法分：方法或公式正確即可給分，即使其後計算有誤
- A：答案分：數值或代數式完全正確方可給分
- PROOF：證明題：推理完整且結論正確方可給分

Section A(1) (35 marks)

1. Factorizing a Quadratic Trinomial and Hence a Five-term Expression

a (2 marks)

M1 $Look for two integers with sum 2 and product -15, namely 5 and -3$ f.t.

f.t. applicable

A1 $x^2+2x-15=(x+5)(x-3)$

b (1 mark)

A1 $x^2+2x-15-4xy-20y=(x+5)(x-3)-4y(x+5)=(x+5)(x-3-4y)$ f.t.

f.t. applicable

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2. Simplifying a Quotient Raised to a Negative Fractional Index and Hence Evaluating It

a (2 marks)

M1 $\left(\frac{8x^{-3}}{27y^6}\right)^{-\frac{1}{3}}=\left(\frac{27y^6}{8x^{-3}}\right)^{\frac{1}{3}}=\left(\frac{27x^3y^6}{8}\right)^{\frac{1}{3}}$ f.t.

f.t. applicable

A1 $=\frac{3xy^2}{2}$

b (1 mark)

A1 $\frac{3(4)\left(\frac{1}{3}\right)^2}{2}=\frac{\frac{4}{3}}{2}=\frac{2}{3}$ f.t.

f.t. applicable

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3. Making the Common Difference the Subject of the Sum Formula of an Arithmetic Series

a (2 marks)

M1 $2S=n[2a+(n-1)d], so (n-1)d=\frac{2S}{n}-2a$ f.t.

f.t. applicable

A1 $d=\frac{2(S-an)}{n(n-1)}$

b (1 mark)

A1 $d=\frac{2[185-5(10)]}{10(9)}=\frac{270}{90}=3$ f.t.

f.t. applicable

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4. Angles of a Kite and the Angles Made by One of Its Diagonals

a (2 marks)

M1 $AC is the axis of symmetry of the kite ABCD, so \angle ADC = \angle ABC = 106°$ f.t.
f.t. applicable

A1 $\angle BCD = 360° - 48° - 106° - 106° = 100°$

b (2 marks)

M1 $AC bisects \angle BCD and \angle BAD$ f.t.
f.t. applicable

A1 $\angle ACD = \frac{100°}{2} = 50° and \angle CAD = \frac{48°}{2} = 24°$

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5. Finding a Principal from Two Years' Compound Interest and Hence the Amount after Three Years

a (2 marks)

M1 $Let the sum be P dollars. Then P(1+5\%)^2 - P = 3075, i.e. 0.1025P = 3075$ f.t.
f.t. applicable

A1 $P = 30000, i.e. the sum of money is 30000 dollars$
u-1 for missing unit

b (2 marks)

M1 $The amount = 30000(1+5\%)^3$ f.t.
f.t. applicable

A1 $= 34728.75 \approx 34729 dollars$
u-1 for missing unit

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6. The Leg and the Area of an Isosceles Triangle from Its Perimeter and Base

a (2 marks)

M1 $AB + AC = 36 - 10 = 26cm and AB = AC$ f.t.
f.t. applicable

A1 $AB = \frac{26}{2} = 13cm$
u-1 for missing unit

b (2 marks)

M1 $Let M be the mid-point of BC. Then AM \perp BC and AM = \sqrt{13^2 - 5^2} = 12cm$ f.t.
f.t. applicable

A1 $The area = \frac{1}{2}(10)(12) = 60\text{cm}^2$
u-1 for missing unit

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7. Finding an Unknown Datum from a Median and Hence the Mean and the Range

a (2 marks)

M1 $The other four data in ascending order are 3, 8, 12, 15, so the median must be x itself$ f.t.
f.t. applicable

A1 $x = 9$

b (2 marks)

A1 $The mean = \frac{3+9+8+12+15}{5} = \frac{47}{5} = 9.4$ f.t.
f.t. applicable

A1 $The range = 15 - 3 = 12$

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8. Forming a Quadratic Equation from a Rectangle and Hence Finding a Diagonal

a (3 marks)

M1 $Let the width be x cm. Then x(x+7) = 120$ f.t.
f.t. applicable

M1 $x^2 + 7x - 120 = 0, i.e. (x+15)(x-8) = 0$ f.t.
f.t. applicable

A1 $x = -15 (rejected) or x = 8, so the width is 8 cm$
u-1 for missing unit

b (2 marks)

M1 $The length = 8 + 7 = 15 cm, so the diagonal = \sqrt{8^2 + 15^2}$ f.t.
f.t. applicable

A1 $= \sqrt{289} = 17 cm$
u-1 for missing unit

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9. The Intersection of Two Lines and Hence a Perpendicular Line through It

a (3 marks)

M1 $From x - y - 1 = 0, x = y + 1$ f.t.
f.t. applicable

M1 $Substituting into 2x + y - 11 = 0 gives 2(y+1) + y - 11 = 0, i.e. 3y = 9$ f.t.
f.t. applicable

A1 $y = 3 and x = 4, so P = (4, 3)$

b (2 marks)

M1 $The slope of L_1 is -2, so the required slope = \frac{1}{2} and the equation is y - 3 = \frac{1}{2}(x - 4)$ f.t.
f.t. applicable

A1 $x - 2y + 2 = 0$

Section A(2) (35 marks)

10. Pressure Varying Directly as Temperature and Inversely as Volume

a (2 marks)

M1 $Let P = \frac{kT}{V}. Then 120 = \frac{k(300)}{5} = 60k$ f.t.
f.t. applicable

A1 $k = 2, so P = \frac{2T}{V}$

b (2 marks)

M1 $P = \frac{2(450)}{6}$ f.t.
f.t. applicable

A1 $= 150$

c (2 marks)

M1 $The new pressure = \frac{2(1.2T)}{0.75V} = 1.6 \times \frac{2T}{V}$ f.t.
f.t. applicable

A1 $So the pressure is increased by 60\%$

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11. Solving a Triangle from Two Base Angles and an Altitude

a (2 marks)

M1 $In\triangle ABD, BD = \dfrac{9}{\tan 52°}; in\triangle ACD, CD = \dfrac{9}{\tan 41°}$ f.t.

f.t. applicable

A1 $BD = 7.03 m and CD = 10.4 m (correct to 3 sig. fig.)$

u-1 for missing unit

b (2 marks)

A1 $BC = BD + CD = 7.0316 + 10.3534 = 17.4 m (correct to 3 sig. fig.)$ f.t.

f.t. applicable

A1 $The area = \dfrac{1}{2}(17.385)(9) = 78.2 \text{m}^2 (correct to 3 sig. fig.)$

u-1 for missing unit

c (3 marks)

M1 $AB = \dfrac{9}{\sin 52°}$ f.t.

f.t. applicable

A1 $AB = 11.4 m (correct to 3 sig. fig.)$

u-1 for missing unit

A1 $Let h be the distance from D to AC. Then h = CD \sin 41° = 10.3534 \sin 41° = 6.79 m$

u-1 for missing unit

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12. An Arithmetic Series Given the Quadratic Formula of Its Partial Sums

a (2 marks)

M1 $The first term = S_1 = 3(1)^2 + 5(1) = 8 and S_2 = 3(2)^2 + 5(2) = 22$ f.t.

f.t. applicable

A1 $The 2nd term = 22 - 8 = 14$

b (2 marks)

A1 $The common difference = 14 - 8 = 6$ f.t.

f.t. applicable

A1 $The nth term = 8 + (n-1)(6) = 6n + 2$

c (3 marks)

M1 $3n^2 + 5n > 1000, so 3n^2 + 5n - 1000 > 0$ f.t.

f.t. applicable

A1 $S_{17} = 952 \le 1000 and S_{18} = 1062 > 1000, so n = 18$

A1 $The 18th term = 6(18) + 2 = 110$ f.t.

f.t. applicable

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13. A Logarithmic Model Relating Earthquake Magnitude and Released Energy

a (2 marks)

M1 $\log E = 4.8 + 1.5(6) = 13.8$ f.t.
f.t. applicable

A1 $E = 10^{13.8} = 6.31 \times 10^{13} joules (correct to 3 sig.\ fig.)$
u-1 for missing unit

b (2 marks)

M1 $\log\left(2 \times 10^{15}\right) = 4.8 + 1.5M, i.e.\ 15.30103 = 4.8 + 1.5M$ f.t.
f.t. applicable

A1 $M = 7.00069 \approx 7.0$

c (3 marks)

M1 $Let the magnitudes be M and M + 2 with energies E_1 and E_2. Then \log E_2 - \log E_1 = 1.5(M+2) - 1.5M = 3$ f.t.
f.t. applicable

M1 $\log \frac{E_2}{E_1} = 3, so \frac{E_2}{E_1} = 10^3 = 1000$ f.t.
f.t. applicable

PROOF $This ratio is independent of M, so the claim is correct and I agree$
pp-1 for poor presentation

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14. A Solid Cylinder with a Cone of the Same Base Removed from Its Top

a (2 marks)

M1 $The volume of the cylinder = \pi(9)^2(12)$ f.t.
f.t. applicable

A1 $= 972\pi \text{cm}^3$
u-1 for missing unit

b (2 marks)

A1 $The cone and the cylinder have the same base and height, so the volume of the cone = \frac{1}{3}(972\pi) = 324\pi \text{cm}^3$ f.t.
f.t. applicable

A1 $The volume of the ornament = 972\pi - 324\pi = 648\pi \text{cm}^3$
u-1 for missing unit

c (2 marks)

A1 $The slant height = \sqrt{9^2 + 12^2} = \sqrt{225} = 15 cm$ f.t.
f.t. applicable

A1 $The curved surface area = \pi(9)(15) = 135\pi \text{cm}^2$
u-1 for missing unit

d (2 marks)

M1 $The total surface area = \pi(9)^2 + 2\pi(9)(12) + 135\pi = 81\pi + 216\pi + 135\pi = 432\pi \text{cm}^2$ f.t.
f.t. applicable

A1 $= 1357.168\ldots \approx 1360 \text{cm}^2 (correct to 3 sig.\ fig.)$
u-1 for missing unit

Section B (35 marks)

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15. Finding Two Real Numbers from a Product of Complex Numbers and Hence a Square

a (3 marks)

M1 $(a+bi)(3-2i) = 3a - 2ai + 3bi - 2bi^2 = (3a+2b) + (3b-2a)i$ f.t.
f.t. applicable

M1 $Comparing the real and imaginary parts, 3a + 2b = 13 and -2a + 3b = 13$ f.t.
f.t. applicable

A1 $a = 1 and b = 5$

b (2 marks)

M1 $(1+5i)^2 = 1 + 10i + 25i^2 = 1 + 10i - 25$ f.t.
f.t. applicable

A1 $= -24 + 10i, so the real part is -24 and the imaginary part is 10$

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16. The Range of a Parameter for Which a Quadratic Equation Has No Real Roots

a (3 marks)

M1 $\textit{Therearenorealroots, sothediscriminant}\Delta = (k-3)^2 - 4(1)(k) < 0$ f.t.
f.t. applicable

M1 $k^2 - 10k + 9 < 0, \textit{i.e.}\ (k-1)(k-9) < 0$ f.t.
f.t. applicable

A1 \
(**b** (2 marks)
1

A1 $\textit{Thegreatestintegralvalueis}8$ f.t.
f.t. applicable

A1 $\textit{Theleastintegralvalueis}2$

17. Simplifying a Difference of Fourth Powers of Sine and Cosine and Hence Solving an Equation

a (2 marks)

M1 $\sin^4\theta - \cos^4\theta = \left(\sin^2\theta - \cos^2\theta\right)\left(\sin^2\theta + \cos^2\theta\right) = \sin^2\theta - \cos^2\theta$ f.t.
f.t. applicable

A1 $= \sin^2\theta - \left(1 - \sin^2\theta\right) = 2\sin^2\theta - 1$

b (2 marks)

M1 $2\left(\frac{3}{5}\right)^2 - 1 = \frac{18}{25} - 1$ f.t.
f.t. applicable

A1 $= -\frac{7}{25}$

c (2 marks)

M1 $2\sin^2\theta - 1 = -\frac{1}{2}, \textit{so}\ \sin^2\theta = \frac{1}{4}, \textit{i.e.}\ \sin\theta = \pm\frac{1}{2}$ f.t.
f.t. applicable

A1 $\theta = 30°, 150°, 210° \textit{or} 330°$
pp-1 for poor presentation

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18. A Circle through Two Points with Its Centre on a Given Line

a (3 marks)

M1 $\textit{Letthecentrebe}(h, k). \textit{Then}\ h - k - 2 = 0, \textit{i.e.}\ k = h - 2$ f.t.
f.t. applicable

M1 $(h-1)^2 + k^2 = (h-8)^2 + (k-7)^2, \textit{whichsimplifiesto}\ 14h + 14k - 112 = 0, \textit{i.e.}\ h + k = 8$ f.t.
f.t. applicable

A1 $h = 5 \textit{and} k = 3, \textit{sothecentreis}(5, 3) \textit{andtheradius} = \sqrt{(5-1)^2 + 3^2} = 5$

b (2 marks)

M1 $(x-5)^2 + (y-3)^2 = 5^2$ f.t.
f.t. applicable

A1 $x^2 + y^2 - 10x - 6y + 9 = 0$

c (2 marks)

M1 $\textit{Thedistancefromthecentretoy} = 6 \textit{is} 6 - 3 = 3, \textit{sohalfthechord} = \sqrt{5^2 - 3^2} = 4$ f.t.
f.t. applicable

A1 $\textit{Thelengthofthechord} = 2(4) = 8$
u-1 for missing unit

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19. Similar Triangles Formed by Two Intersecting Chords and the Shortest Chord through a Point

a (3 marks)

M1 $\angle RPE = \angle QSE (angles in the same segment on arc RQ)$ f.t.

f.t. applicable

M1 $\angle PER = \angle SEQ (vert.\ opp.\ angles)$ f.t.

f.t. applicable

PROOF $Hence \triangle PER \sim \triangle SEQ (A.A.A.)$

pp-1 for poor presentation

b (9 marks)

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